

Distinct Sums Grid

Time limit: 1.00 s

Memory limit: 512 MB

Create an $n \times n$ grid that fulfills the following requirements:

1. Each integer $1 \dots n$ appears n times in the grid.
2. If we create a set that consists of all sums in rows and columns, there are $2n$ distinct values.

Input

The only line has an integer n .

Output

Print a grid that fulfills the requirements. You can print any valid solution. If there are no solutions, print IMPOSSIBLE.

Constraints

- $1 \leq n \leq 1000$

Example

Input	Output
5	2 3 1 1 1 1 5 5 3 3 2 3 5 2 4 5 4 5 4 1 2 3 4 4 2

Explanation: Each integer $1 \dots 5$ appears 5 times, and the sums in rows and columns are $\{8, 11, 12, 14, 15, 16, 17, 18, 19, 20\}$.